

أسئلة سنوات سابقة

نظرية تمكم

ابذل غاية الجهد ييسر لك غاية الهدى
(والذين جاهدوا فينا لنهدينهم سبلنا)

عضو الجمعية الطلابية:
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Q1) Select the most accurate answer for each of the following statements: (30 marks)

1) The inverse Laplace transforms of $F(s) = \frac{2s+3}{s^2+4s+13}$ is:

- A) $f(t) = e^{-2t} (2 \cos 3t - \frac{1}{3} \sin 3t)$ B) $F(s) = e^{-2t} (2 \cos 3t - \sin 3t)$ C) $f(t) = e^{-2t} (2 \cos 3t - \frac{1}{3} \sin 3t)$
D) None.

2) In open loop control:

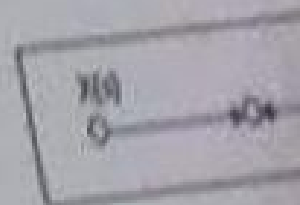
- A) Only system parameters have effect on the controlled variable
B) Output is dependent on control input C) only feedback signal effect the output D) None.

3) The poles of $Y(s)$ obtained from $y'' + by' + 4y = 0$, and $y(0) = 1, y'(0) = 0$ are complex conjugate if

- A) $0 < b < 4$ B) $b > 4$ C) $b = 4$ D) none

4) The input is $x(t)$ and the output is $c(t)$, the shown figure represents a

- A) branch B) forward path C) loop D) none



5) The output of the comparator in a closed loop control system is called

- A) error signal B) Feedback signal C) output signal. D) Difference signal

6) In a temperature control system, what represents the output of the system?

- a) The required temperature b) The heat produced by the system.
c) The actual temperature achieved d) The heating element.

7) The major disadvantage of a feedback system may be

- (a) instability (b) inefficiency (c) Unreliability (d) Inaccuracy

8) For stability analysis, the s -contour should cover

- A) the right hand side of s -plane C) the left hand side of s -plane B) the
D) the lower side of s -plane.

- 9) The control system given as a characteristic equation $s^2 + 4s^2 + s + b = 0$ is stable if
 A) $0 < b < 4$ B) $b > 4$ C) $b < 4$ D) $b = 4$
- 10) Adding PI controller to a system equivalent to
 A) real zero and pole at the origin B) pole at the origin C) real zero and real pole D) two real poles
- 11) The system with a characteristic equation $s^3 + 4s^2 + 4s + K = 0$ has two imaginary conjugate poles if
 A) $K > 4$ B) $K < 4$ C) $K < 16$ D) $K = 16$

A second order system has settling time 4 sec for 2% error and peak time 3.14 sec. Answer 12 and 13

- 12) The damping constant ($\zeta \omega_n$) of this system is
 A) 2 B) 1 C) 0 D) none
- 13) The undamped natural frequency is
 A) 0 B) 1 C) 2 D) none

- 14) An open loop transfer function has two poles $p_1 = 1$ and $p_2 = -1$; the closed loop system is stable if the number of encirclements of Nyquist plot off $(-1, j0)$ point is
 A) -1 B) 1 C) 0 D) none

A control system has three state variables, two inputs and two outputs, the dimension of the matrix C is

- A) 3×2 B) 3×3 C) 2×2 D) 2×3



Q2) Plot an approximate Bode diagram for $GH(s) = \frac{s(0.1s+1)}{(10s+1)}$; show on the plot the slope of each part and the corner frequencies. (10 marks)

Q3) Plot a root locus using $GH(s) = \frac{s+2}{(s+10)(s^2+8s+32)}$, show on the plot specific points (10 marks)

Choose the correct answer using MATLAB.

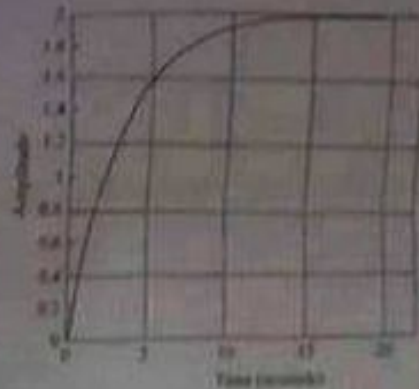
Q1) Closed-loop transfer function of a unity-feedback system is given by $Y(s)/R(s) = 1/(s\tau + 1)$. Steady-state error to unit-ramp input is

- (A) ∞ (B) τ (C) 1 (D) $1/\tau$

Q2)

The plot in Fig. shows the unit step response of a first-order system. Transfer function of the system is

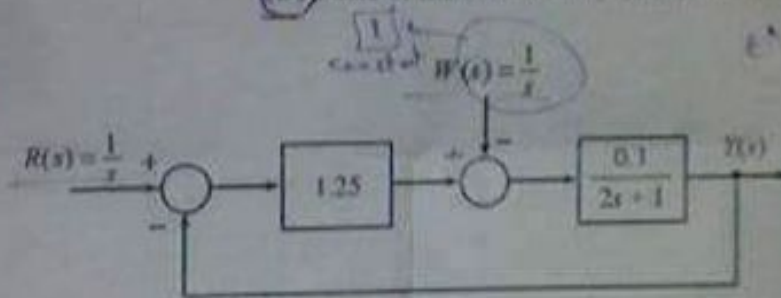
- (A) $\frac{1}{5s+1}$ (B) $\frac{2}{15s+1}$
(C) $\frac{2}{3s+1}$ (D) $\frac{1}{15s+1}$



Q3)

The system of Fig. has the steady-state error $e_{ss} = ?$

- (A) $\frac{1}{1.125}$ (B) $\frac{0.1}{1.125}$
(C) $\frac{1.1}{1.125}$ (D) None of the answers in (A), (B), and (C) is correct.

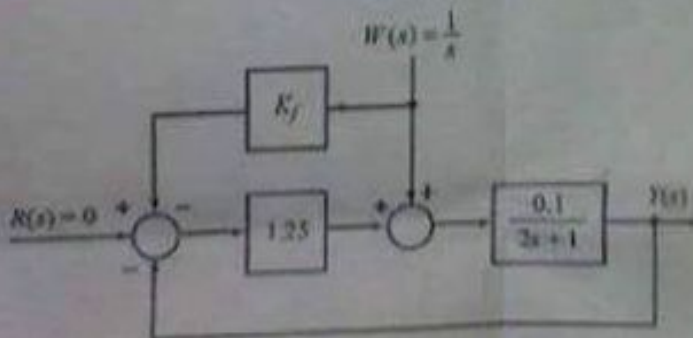


Handwritten calculation:
 $e_{ss} = \lim_{s \rightarrow 0} s E(s)$
 $= 1 - C_{ss}$
 $= 1 - 0.94 = 0.06$

Q4)

Consider the system of Fig. The steady-state error is zero for

- (A) $K_f = 0$ (B) $K_f = \frac{1}{1.25}$
(C) $K_f = 1.25$ (D) None of the answers in (A), (B), and (C) is correct





Student ID: 3115018023

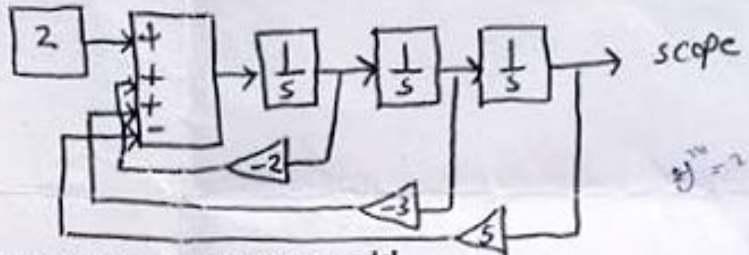
mid Exam
20 points

اسم الطالب: أشرف خالد الشيش

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1. The differential equation that can be solved by the following model is

- a. $\ddot{y}(t) + 2\dot{y}(t) + 3y(t) - 5y(t) = 2$
b. $\ddot{y}(t) + 2\dot{y}(t) + 5y(t) + 3y(t) = 2$
c. $2\ddot{y}(t) - 2\dot{y}(t) - 3y(t) + 5y(t) = 2$
d. $\ddot{y}(t) + 2\dot{y}(t) + 3y(t) + 5y(t) = 2$



2. If we represent the following differential equation using state space model
 $\ddot{y}(t) + 3\dot{y}(t) + 5y(t) + 2y(t) = r(t)$, $r(t) = 2$
The matrix A =

a. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & -3 \end{bmatrix}$

b. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -5 & -3 \end{bmatrix}$

c. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 5 & 3 \end{bmatrix}$

d. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & -5 & -2 \end{bmatrix}$

e. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -5 & -2 \end{bmatrix}$

3. The transfer function that represent this differential equation
 $\ddot{y}(t) + 3\dot{y}(t) + 5y(t) + 2y(t) = r(t)$, $r(t) = 2$

a. $\frac{2}{s(s^3+3s^2+5s+2)}$

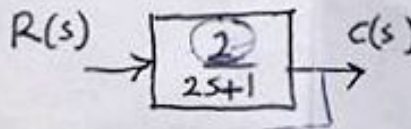
b. $\frac{1}{s(s^3+3s^2+5s+2)}$

c. $\frac{1}{(s^3+3s^2+2s+5)}$

d. $\frac{1}{(s^3+3s^2+5s+2)}$

e. $\frac{2}{(s^3+3s^2+5s+2)}$

***for the following system



4. The time required to reach the steady state value =
a. 2-3 sec b. 4-5 sec c. 6-8 sec

d. 8-10 sec

5. If $r(t) = 2$, the final value =
a. 1 b. 2 c. 3

d. 4

***if a negative feedback $H(s) = 2$ was added, answer the following

6. The total transfer function become =

a. $\frac{2}{2s+2}$

b. $\frac{4}{2s+1}$

c. $\frac{2}{2s+5}$

d. $\frac{2}{2s+4}$

e. $\frac{1}{s+2}$

7. The gain of the system
a. Increased b. decreased c. not effected

8. The system become
a. Faster b. slower c. not effected

9. The final value of the system when $r(t) = 3$ is
a. 1 b. 3 c. 12

d. 1.2

*** To represent the transfer function $\frac{s+1}{s^2+2}$

in mat lab Simulink, the numerator matrix = [1 1]

The denominator matrix = [1 0 2]

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1	2	3	4	5	6	7	8	9
d	e	d	d	d	c	b	c	d

$\frac{3}{5} = \frac{6}{5} = \frac{12}{10} = 1.2$

$C(s) = 2$

$$S(s^2+3s^2+5s+2)$$

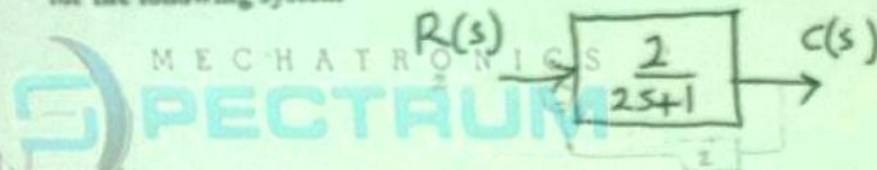
$$S(s^2+3s^2+5s+2)$$

$$(s^2+3s^2+2s+5)$$

$$(s^2+3s^2+5s+2)$$

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***for the following system



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$$\frac{2}{2s+1}$$

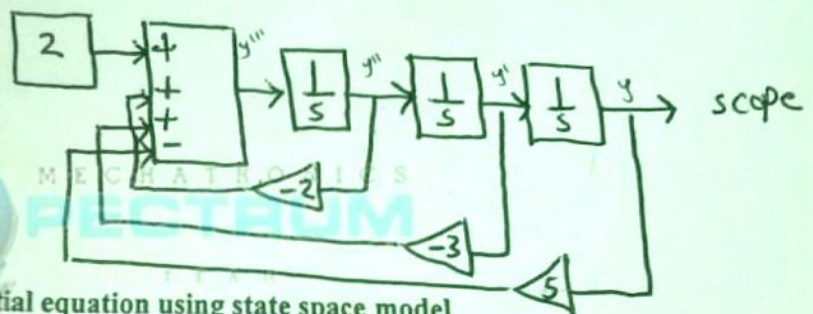
$$\frac{2}{2s+1} + \frac{4}{2s+1} = \frac{6}{2s+1}$$

$$\frac{2}{2s+5}$$

$$\frac{2}{2s+1} + \frac{4}{2s+1} = \frac{6}{2s+1}$$

1. The differential equation that can be solved by the following model is

- a. $\ddot{Y}(t) + 2\dot{y}(t) + 3y(t) - 5y(t) = 2$
- b. $\ddot{Y}(t) + 2\dot{y}(t) + 5\dot{y}(t) + 3y(t) = 2$
- c. $2\ddot{Y}(t) - 2\dot{y}(t) - 3\dot{y}(t) + 5y(t) = 2$
- d. $\ddot{Y}(t) + 2\dot{y}(t) + 3\dot{y}(t) + 5y(t) = 2$



2. If we represent the following differential equation using state space model
 $\ddot{Y}(t) + 3\dot{y}(t) + 5\dot{y}(t) + 2y(t) = r(t)$, $r(t) = 2$

The matrix A =

- a. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & -3 \end{bmatrix}$
- b. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -5 & -3 \end{bmatrix}$
- c. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 5 & 3 \end{bmatrix}$
- d. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & -5 & -2 \end{bmatrix}$
- e. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -5 & -2 \end{bmatrix}$

3. The transfer function that represent this differential equation

$$\ddot{Y}(t) + 3\dot{y}(t) + 5\dot{y}(t) + 2y(t) = r(t), r(t) = 2$$

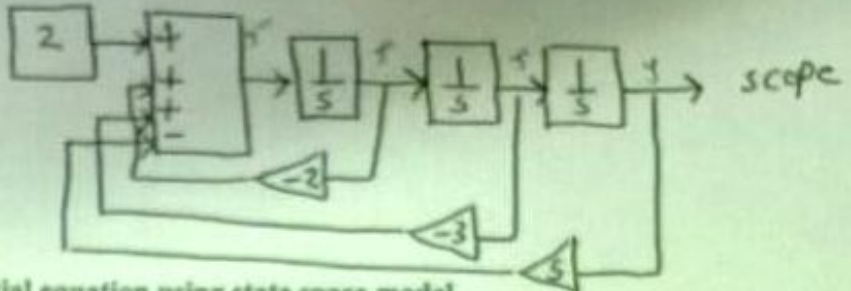
- a. $\frac{2}{s(s^2+3s^2+5s+2)}$
- b. $\frac{1}{s(s^2+3s^2+5s+2)}$
- c. $\frac{1}{(s^2+3s^2+2s+5)}$
- d. $\frac{1}{(s^3+3s^2+5s+2)}$
- e. $\frac{2}{(s^3+3s^2+5s+2)}$

***for the following system

$P(s)$

1. The differential equation that can be solved by the following model is

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 b. $\ddot{y}(t) + 2\dot{y}(t) + 5y(t) + 3y(t) = 2$
 c. $\ddot{y}(t) - 2\dot{y}(t) - 3y(t) + 5y(t) = 2$
 d. $\ddot{y}(t) + 2\dot{y}(t) + 3y(t) + 5y(t) = 2$



2. If we represent the following differential equation using state space model

$$\ddot{y}(t) + 3\dot{y}(t) + 5y(t) + 2y(t) = r(t), \quad r(t) = 2$$

The matrix A =

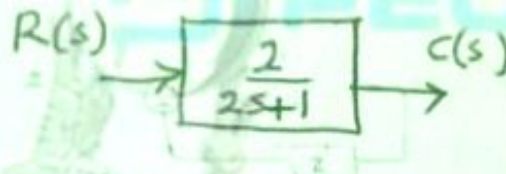
- a. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & -3 \end{bmatrix}$ b. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -5 & -3 \end{bmatrix}$ c. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 5 & 3 \end{bmatrix}$ d. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & -5 & -2 \end{bmatrix}$ e. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -5 & -2 \end{bmatrix}$

3. The transfer function that represent this differential equation

$$\ddot{y}(t) + 3\dot{y}(t) + 5y(t) + 2y(t) = r(t), \quad r(t) = 2$$

- a. $\frac{2}{s(s^2+3s^2+5s+2)}$ b. $\frac{1}{s(s^2+3s^2+5s+2)}$ c. $\frac{1}{(s^2+3s^2+2s+5)}$ d. $\frac{1}{(s^2+3s^2+5s+2)}$ e. $\frac{2}{(s^2+3s^2+5s+2)}$

for the following system



The time required to reach the steady state value =

- a. 2-3 sec b. 4-5 sec c. 6-8 sec d. 8-10 sec

If $r(t) = 2$, the final value =

- a. 1 b. 2 c. 3 d. 4

negative feedback $H(s) = 2$ was added, answer the following

- a. $\frac{2}{2s+2}$ b. $\frac{4}{2s+1}$ c. $\frac{2}{2s+5}$ d. $\frac{2}{2s+4}$ e. $\frac{1}{s+3}$

gain of the system

Increased decreased c. not effected

system become

faster b. slower c. not effected

final value of the system when $r(t) = 3$ is

- a. 3 b. 12 c. 12 d. 1.2

at the transfer function $\frac{s+1}{s^2+2}$

in mat lab Simulink, the numerator matrix =

$$\text{Numerator} = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

matrix = $\begin{bmatrix} 1 & 0 & 2 \end{bmatrix}$



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$$S(s^2+3s^2+5s+2)$$

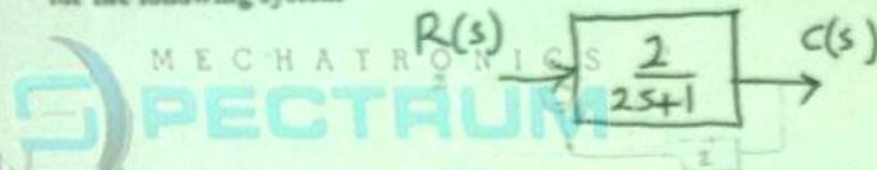
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*** To represent the transfer function $\frac{s+1}{s^2+2}$ in mat lab Simulink, the numerator matrix = $\begin{bmatrix} 1 & 1 \end{bmatrix}$,

The denominator matrix = $\begin{bmatrix} 1 & 0 & 2 \end{bmatrix}$

Numerator = $\begin{bmatrix} 1 & 1 \end{bmatrix}$

$$\frac{2}{2s+1} = \frac{2}{1+2s} = \frac{2}{2s+1}$$

$$\frac{2}{2s+1} = \frac{2}{1+2s} = \frac{2}{2s+1}$$

$$\frac{2}{2s+1} = \frac{2}{1+2s} = \frac{2}{2s+1}$$

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